HCC1395: somatic mutation and RNA evidence

Whole source-call audit | matched-normal DNA baseline | original tumor RNA | pinned reference ORFs. Counts below describe the specified public call sets and RNA run, not every biological mutation or a validated full-length isoform.

At a glance

Evidence layer	Count	Interpretation
Ingested small variants	41,072	39,447 SNVs; 1,625 indels
Ingested SV source records	1,788	DNA breakpoint records/events, counted in their source sets
Ensembl 87 coding effect	491	Any annotated protein-coding transcript, reference model
orfdata ncORF coding-block overlap	17	Candidate reference ORF overlap; distinct from mutant translation
Original RNA ALT-observed calls	343	2,681 ALT reads; 2,681 ALT fragments
Calls with local mutant protein contexts	22	119 ranked contexts; full-length isoforms unresolved

Coding annotation and its denominator

Of 41,072 DNA small-variant records, 491 change the coding sequence of at least one Ensembl 87 transcript. The pinned orfdata 0.34.1 GRCh38 catalog (7,264 reference ncORF models; release 2026-10-11.1) has exact coding-block overlaps at 17 calls. The groups overlap at 3 calls, so the union is 505. These are annotation opportunities. The catalog does not assess a cell-line-specific mutant ORF, RNA expression, or peptide evidence.

Reference annotation	All DNA calls	With original RNA ALT
Ensembl coding effect	491	135
orfdata coding-block overlap	17	5
Union of either category	505	see per-call table

SNVs query their changed reference base. Deletions query removed reference bases; insertions use the VCF anchor as a candidate overlap. The per-call table preserves that rule. A coding-block hit does not prove the inserted sequence is in frame.

Original RNA and local protein evidence

RNA assay: SRR23456709, ONT direct RNA. Each nominated small variant has a status from this run. An ALT read is a sequence observation; it is not a full-length mutant isoform.

RNA allele status	Small-variant records
alternate observed	343
reference only	239
reference and other only	53
other allele only	19
alignment without allele call	7,300
no alignment overlap	33,118

The 343 ALT-observed calls contribute 2,681 allele-read and 2,681 fragment observations. Isovar resolves at least one read-compatible **local** mutant protein context for 22 calls (119 ranked rows). The remaining 321 ALT-observed calls have no local translation under the stated method. No-overlap and uncallable statuses cannot establish absence of RNA expression.

Representative mutant residues with original-read support

SEQC2 call	Top local amino-acid context	Mutant interval	Supporting reads
OSTC F9L / COSM32736	METLYRVPLLVLECPNLKLKKPPWL	[8,9)	5
PRDX5 F157L / COSM32759	VRLLADPTGALGKETDLLLD	[10,11)	6

The SEQC2 tumor/normal DNA VAFs are 0.490/0.001 (OSTC) and 0.442/0.002 (PRDX5). An independent proteomic study observed the mutant residues, not these entire local sequence contexts. A separate first-100,000-read ONT cDNA fixture has a PRDX5 local context with six supporting reads; its bounded counts are not included in the whole-run denominators.

Structural-variant RNA screen

Breakend screen outcome	Source records
No breakend-window alignment	1,233
One end covered; no split candidate	149
Both ends covered; no split candidate	400
Split candidate; path unresolved	6

Six Talsania records have same-read split-alignment candidates (17 record-level observations). Published POU2F1-NME7 has one candidate read. Junction bases, orientation, splice choice and ORF remain unresolved; no fusion protein is called.

Sources, processing and limits

Input	Pinned scope
Matched-normal small variants	SEQC2 v1.2.1 SNV + indel, GRCh38.d1.vd1; HCC1395BL normal
SV benchmark	Talsania et al. 2022 Table S4, HCC1395 vs HCC1395BL
Original RNA	PRJNA934933 / SRR23456709; 1,049,452 ONT direct RNA reads
Alignment	minimap2 2.26 -ax splice -uf -k14; NCBI GRCh38 no-alt

Reference model and interpretation

Small-variant coding effects use Varcode with Ensembl release 87 on GRCh38. The human orfdata 0.34.1 catalog uses Ensembl 101 GRCh38 reference ORFs (7,264 models, dataset 2026-10-11.1). An orfdata coding-block overlap is a reference annotation candidate; this analysis did not translate mutated ncORFs from the RNA reads. Genomic transcript overlap is not RNA-compatible isoform support. Original-read allele counts and local protein contexts are separate observations.

SHA-256 input pins

SEQC2 SNV VCF gzip

`0c49f7e1519fc7860b773f79bdcf07453fe4def8303707f34542471d92494c9d`
SEQC2 indel VCF gzip

`db7dc8be2ec04bab830ff1411146c7d538c217ff684e874b85d98a3dbbda91cf`
Talsania Table S4 xlsx

`11e8d6393b83f0e0b7fdb0b13a57376663fc9662116d78c89a0d5fa530df096b`
SRR23456709 SRA archive

`d69db8e48bbb20665c3e03a1fa35742be818a0127994e015cb3a560afae39ac8`
GRCh38 no-alt FASTA (uncompressed)

`9cce8b926416dd96b152deea85188495b75f7ac8d634cc723a017067be8702b7`
orfdata bundled GRCh38 catalog

`06bc0bf3f81c25a7356f71a38581ff5704c05dcb75752454bc1ca75075ab8c09`

What remains unresolved

ONT raw-read errors can create other-base calls and competing local contexts. The RNA and DNA libraries come from separate studies. A 500-base breakend window and split-alignment candidate are screening evidence only. Full-length isoform linkage and SV protein reconstruction were not assessed.

Machine-readable per-call annotation, allele, protein, and SV tables and complete reconstruction commands are published with the Isovar HCC1395/COLO829 reports. These PDFs summarize selected public source call sets, not a complete catalogue of every cell-line mutation.